PREDICTIVE ANALYTICS FOR AIR QUALITY CATEGORY CLASSIFICATION USING MACHINE LEARNING

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***Abstract*—**Air quality is influenced by particulate matter, nitrogen oxides, sulphur dioxide, volatile organic compounds, carbon gases, methane, and temperature. This paper presents a supervised machine-learning workflow for air-quality category prediction using a supplied dataset containing 500 records and 10 columns. The dataset has no label, so the target AIR_QUALITY is derived from PM2.5 using CPCB-style bands (Good, Satisfactory, Moderate, Poor); PM2.5 is excluded and nine numerical variables are used as inputs. The workflow includes data-quality inspection, exploratory data analysis, preprocessing, an 80:20 stratified train-test split, and four classifiers: K-Nearest Neighbors (KNN), Decision Tree, Support Vector Machine (SVM), and Random Forest. SVM achieved the highest accuracy (0.7000), followed by KNN (0.6900), Random Forest (0.6700), and Decision Tree (0.5200) on the held-out split. These results describe performance on the supplied dataset and do not establish real-world forecasting effectiveness.

Keywords—air quality, predictive analytics, machine learning, classification, exploratory data analysis, PM2.5.

I. INTRODUCTION

Air pollution is an important environmental concern in India, and air quality is affected by particulate matter, nitrogen oxides, sulphur dioxide, volatile organic compounds, carbon and methane gases, and temperature. Predictive analytics uses historical data and machine-learning methods to identify patterns and estimate likely outcomes. When the outcome is an air-quality class, the task can be formulated as supervised multiclass classification.

The purpose of this work is to apply the Predictive Analytics course workflow to an air-quality dataset, explore relationships among recorded variables, train multiple classifiers, and compare their performance. The study is educational and does not present the model as a substitute for calibrated monitoring stations.

A. Objectives and Scope

The objectives are to analyze relationships between pollutant readings and air-quality classes; develop classifiers using gas and particulate inputs; compare KNN, Decision Tree, SVM, and Random Forest; and interpret predictions through confusion matrices. The scope covers data inspection, EDA, target construction, feature selection, numerical scaling, model training, and evaluation. Production deployment, live sensor integration, pollution forecasting, and guaranteed health advice are outside scope.

II. DATASET DESCRIPTION

The project uses the file air_quality_dataset.csv supplied as project input. The file contains 500 rows and 10 columns. A formal publisher, collection date, location, sampling methodology, and measurement units were not provided; therefore, the dataset is not described here as an official monitoring dataset.

Attribute	Type	Role
PM10, NOx, NO2, SO2	Numeric	Model inputs
VOCs, CO, CO2, CH4, Temperature	Numeric	Model inputs
PM2.5	Numeric	Excluded; defines target
AIR_QUALITY	Categorical	Prediction target

TABLE I. DATASET ATTRIBUTES

The dataset contains 4 air-quality classes after labeling. Pandas checks reported zero missing cells and zero exact duplicate rows. Class counts are imbalanced: Satisfactory 282, Moderate 124, Good 48, and Poor 46. Four readings (one NOx, one NO2, two SO2) are slightly negative, likely sensor noise; they were retained.

The target y is AIR_QUALITY, created from PM2.5 using CPCB-style bands of 0–30 (Good), 31–60 (Satisfactory), 61–90 (Moderate), and above 90 (Poor), assuming µg/m³. One record above 120 was merged into Poor so every class supports a stratified split. The input matrix X contains PM10, NOx, NO2, SO2, VOCs, CO, CO2, CH4, and Temperature. PM2.5 is excluded because the label is a direct function of it and would cause target leakage. All inputs are numeric, so no encoding was needed.

III. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed before modeling to understand data types, distributions, class frequencies, correlations, and group-wise variation. The workflow inspected dataset shape, column names, summary statistics, missing values, duplicates, and negative readings. Visualizations included a class count plot, numerical histograms, a Pearson correlation heatmap, class-wise boxplots, and a class-wise standardized mean heatmap.

A. Air-Quality Class and Numerical Feature Distributions

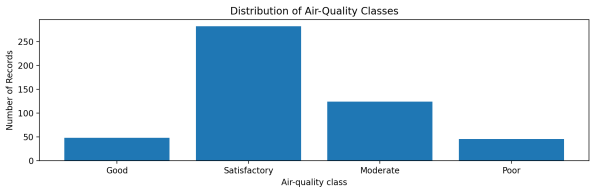


Fig. 1. Distribution of air-quality classes.

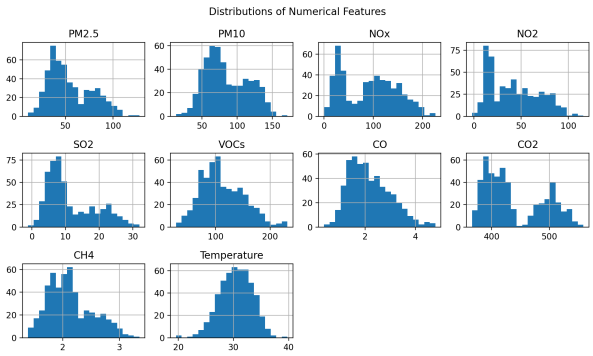


Fig. 2. Histograms of numerical dataset features.

The class distribution analysis found 282 Satisfactory, 124 Moderate, 48 Good, and 46 Poor records, so the classes are imbalanced. Histograms were used to inspect the ranges, concentration, skewness, and possible unusual values of numerical fields. The plots also motivated scaling because numerical features have different magnitudes.

B. Correlation and Pollutant Levels

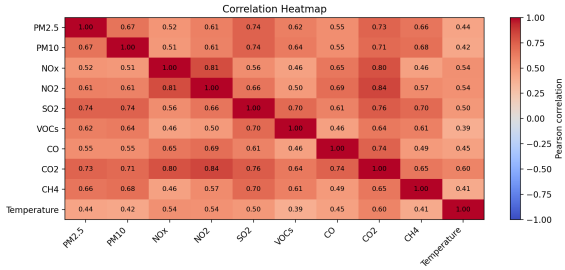


Fig. 3. Correlation heatmap of numerical features.

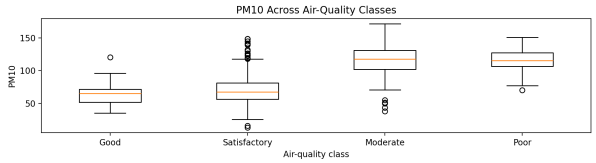


Fig. 4. PM10 values grouped by air-quality class.

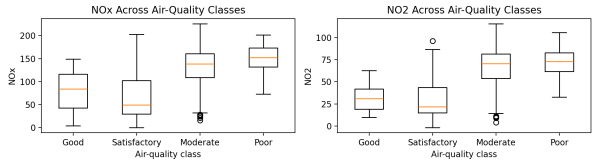


Fig. 5. NOx and NO2 values grouped by air-quality class.

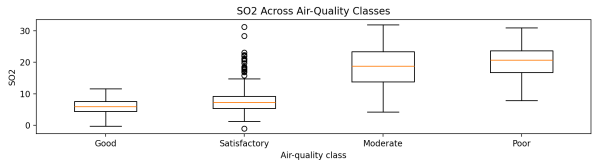


Fig. 6. SO2 values grouped by air-quality class.

Pearson correlation was used to summarize linear association between pairs of numerical fields. The coefficient ranges from −1 to +1; correlation does not imply causation, and low linear correlation does not prove that a feature is unimportant to a nonlinear classifier. All variables are positively correlated; PM2.5 correlates most with SO2 (0.74) and CO2 (0.73), while NO2 and CO2 (0.84) are strongly collinear. PM10, NOx, NO2, and SO2 were compared across air-quality classes using boxplots; these plots describe recorded values and do not establish health thresholds.

C. Gas, Climate and Class-wise Analysis

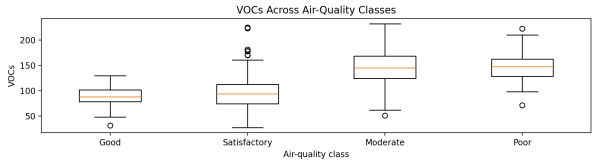


Fig. 7. VOCs values grouped by air-quality class.

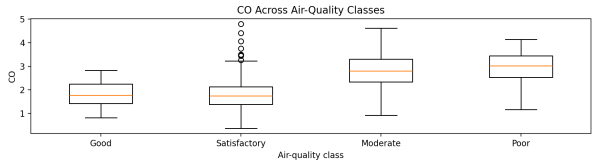


Fig. 8. CO values grouped by air-quality class.

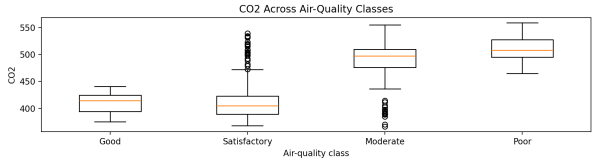


Fig. 9. CO2 values grouped by air-quality class.

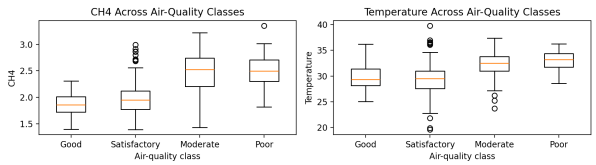


Fig. 10. CH4 and temperature values grouped by air-quality class.

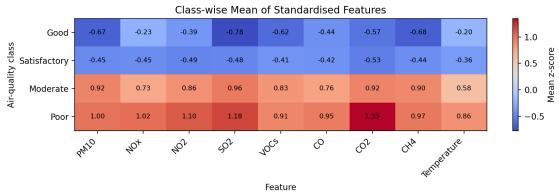


Fig. 11. Class-wise mean of standardized features.

VOCs, CO, CO2, CH4, and temperature were compared across air-quality classes using boxplots, and class-wise means were computed using pandas groupby(). These plots help identify patterns in the supplied records, but units and collection protocols were not documented. Because PM2.5 is only moderately correlated with the other features (0.44–0.74), adjacent classes overlap and may limit accuracy.

IV. CLASSIFICATION METHODOLOGY

The dataset was divided into training and testing subsets using an 80:20 stratified split with random_state=42 (400 training and 100 test records). Stratification preserves relative class frequencies. Preprocessing was fitted on the training subset and then applied to the test subset to reduce information leakage.

The numerical pipeline used median imputation followed by StandardScaler. Although the supplied dataset had no missing values, the imputer was retained as a safeguard for future data changes. No categorical pipeline was needed.

A. K-Nearest Neighbors

KNN predicts a class from the labels of the k closest training observations. Euclidean distance is used for numeric vectors: $d(x,z) = \sqrt{\Sigma_j (x_j - z_j)^2}$. The baseline setting was k = 5. Numerical scaling is important because distance calculations are sensitive to feature magnitude.

B. Decision Tree

A Decision Tree applies sequential feature-based rules. A common split criterion is Gini impurity, defined as $\text{Gini} = 1 - \Sigma_k p_k^2$, where p_k is the proportion of class k at a node. The classifier used random_state=42. Scaling is generally not required by tree models, although the shared preprocessing pipeline was retained.

C. Support Vector Machine

SVM identifies a decision boundary separating classes. The RBF kernel was used: $K(x,z) = \exp(-\gamma\|x - z\|^2)$, where γ controls the influence of individual training examples. Numerical scaling is important for SVM.

D. Random Forest

Random Forest is an ensemble of decision trees trained with randomized samples and feature selection. Predictions are combined by voting. The classifier used 200 trees and random_state=42. The same input fields and train-test split were used for all four models.

V. RESULTS AND DISCUSSION

Accuracy is the fraction of test examples classified correctly. Precision is TP/(TP+FP), recall is TP/(TP+FN), and F1-score is the harmonic mean of precision and recall. Weighted averages combine per-class scores according to the number of true examples in each class. A confusion matrix compares actual and predicted labels; diagonal entries represent correct predictions and off-diagonal entries represent misclassifications.

Classifier	Accuracy	Weighted P	Weighted R	Weighted F1
KNN	0.6900	0.6371	0.6900	0.6467
Decision Tree	0.5200	0.5317	0.5200	0.5219
SVM (RBF)	0.7000	0.5736	0.7000	0.6295
Random Forest	0.6700	0.5694	0.6700	0.6152

TABLE II. CLASSIFICATION PERFORMANCE ON THE HELD-OUT TEST SET

SVM achieved the highest test accuracy (0.7000), followed by KNN (0.6900), Random Forest (0.6700), and Decision Tree (0.5200). The corresponding weighted precision, recall, and F1-scores are shown in Table II. Confusion matrices were generated for all four models using the same class order. Their diagonal cells summarize correct predictions, while off-diagonal cells identify class confusions.

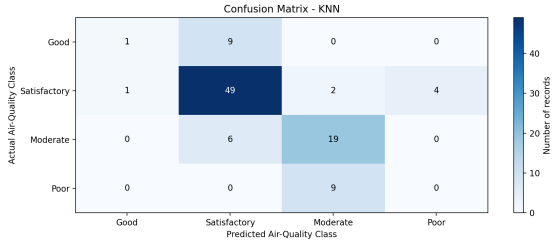


Fig. 12. Confusion matrix for K-Nearest Neighbors (KNN).

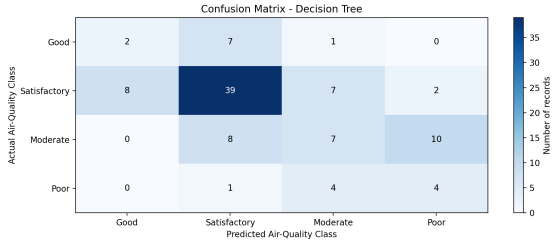


Fig. 13. Confusion matrix for the Decision Tree classifier.

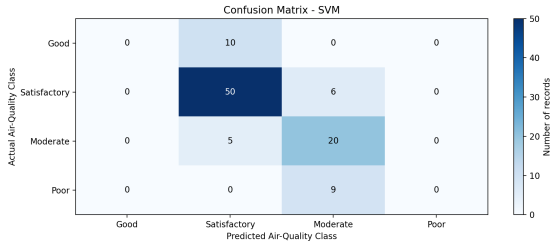


Fig. 14. Confusion matrix for the Support Vector Machine (SVM).

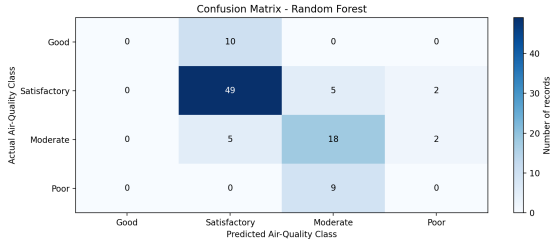


Fig. 15. Confusion matrix for the Random Forest classifier.

Most correct predictions fall in the Satisfactory class (49–50 of 56 for KNN, SVM, and Random Forest) and the Moderate class (18–20 of 25). SVM and Random Forest recognized none of the 10 Good or 9 Poor test records, so their accuracy mainly reflects the majority classes; errors fall between adjacent bands. The Decision Tree was more balanced on minority classes (2 of 10 Good, 4 of 9 Poor) but less accurate overall. Macro F1-scores were low (0.36–0.42). The scores apply only to the supplied dataset, split, and model settings; because the provenance is unknown, they should not be interpreted as proof of performance on new stations, cities, or seasons.

VI. CONCLUSION

This project applied a supervised machine-learning workflow to air-quality category prediction using particulate, gas, and temperature features. Data inspection and EDA were followed by target construction, feature selection, preprocessing, a stratified 80:20 split, and evaluation of KNN, Decision Tree, SVM, and Random Forest. SVM achieved the highest accuracy (0.7000) on the held-out test subset, followed by KNN (0.6900), Random Forest (0.6700), and Decision Tree (0.5200). The work demonstrates classification and confusion-matrix analysis within the educational scope of INT234. It does not include a deployed application and does not claim that model predictions replace official air-quality monitoring.

VII. FUTURE SCOPE

Future work should verify dataset provenance, collection dates, units, locations, sampling approach, and licensing; test the models on independently collected data from different locations and seasons; use stratified cross-validation and training-only hyperparameter tuning; address class imbalance using class weights or resampling; and explore justified feature engineering. Feature-importance methods could help interpret model behavior, but should not be treated as causal evidence. Expert review and a responsible interface that communicates uncertainty would be necessary before practical use.

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APPENDIX: REPRODUCIBILITY NOTES

The reported experiment used random_state=42 and stratification by AIR_QUALITY. Inputs were PM10, NOx, NO2, SO2, VOCs, CO, CO2, CH4, and Temperature. AIR_QUALITY was the target, derived from PM2.5 using bands of ≤30, 31–60, 61–90, and >90, and PM2.5 was excluded. The numerical pipeline used median imputation and StandardScaler. Classifiers were KNeighborsClassifier(n_neighbors=5), DecisionTreeClassifier(random_state=42), SVC(kernel='rbf'), and RandomForestClassifier(n_estimators=200, random_state=42). Preserve the exact dataset and software versions to reproduce the reported metrics.